EXPERIMENT 9

# User Interface Prototype for a GSM Service

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| **Subject** | Mobile Computing |
| **Tool** | Figma |
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## Aim

To design a user interface prototype in Figma for a GSM service covering voice calling, SMS and SIM/network settings.

## Tools Required

- Figma (web or desktop)
- Mobile frame preset (iPhone or Android, 390 x 844)
- Components, Variants and Auto Layout
- Prototype mode with Smart Animate and overlays

## Theory

GSM (Global System for Mobile Communications) is a digital, circuit-switched cellular standard that provides voice calls, SMS, supplementary services and low-rate data. Its architecture has the Mobile Station (ME + SIM), Base Station Subsystem (BTS, BSC) and Network Switching Subsystem (MSC, HLR, VLR, AuC, EIR).

The user interface of a GSM service exposes these functions to the subscriber: dialling, call log, messaging, SIM and network selection, roaming and supplementary services such as call forwarding and call waiting.

- Core screens: Dialer, Call in progress, Call log, Messages, Conversation, SIM and Network settings.
- Status bar shows signal bars, network type, roaming indicator and battery.
- Supplementary services: call forwarding, call barring, call waiting, caller ID.
- SIM details: operator name, IMSI-linked number, PIN lock, network mode.

## Design Specification

| Element | Specification |
|---|---|
| Frame | 390 x 844 mobile frames, 16 px side padding |
| Palette | Blue #1F4E79 header, green #3A9E6B call button, red #D64545 end-call button |
| Typography | Inter: 20 px header, 16 px body, 12 px status text |

| Element | Specification |
|---|---|
| Components | Keypad button, list item, message bubble, toggle, bottom navigation |
| Screens | Dialer, Calling, Call log, Messages, Chat, SIM and Network |

## Procedure in Figma

| Step | Action | Details |
|---|---|---|
| 1 | Create frames | Add a mobile frame and duplicate it for each of the six screens. Name them clearly. |
| 2 | Design the status bar | Create a component with signal bars, 'GSM/4G' label and battery. Reuse it at the top of each screen. |
| 3 | Build the dialer | Make a keypad button component (64 px circle). Use a 3 x 4 Auto Layout grid. Add the number display and a green Call button. |
| 4 | Design the Calling screen | Show contact name, timer, and mute, speaker and keypad buttons with a red End button. |
| 5 | Build Messages | Create message bubble components (sent and received variants) and a text input bar. |
| 6 | Build SIM and Network settings | Create list items with toggles for Roaming, Network mode and Call forwarding. |
| 7 | Add bottom navigation | Create a four-tab component (Calls, Messages, Contacts, Settings) with active and inactive variants. |
| 8 | Connect the prototype | In Prototype mode link Call to Calling screen, End to Dialer, tab icons to their screens, and the Roaming toggle to an overlay warning about charges. |
| 9 | Test | Present the prototype on desktop or the Figma mobile app and fix any dead ends. |

## Prototype Interactions

- Pressing Call opens the Calling screen with a running timer.
- Bottom navigation switches between screens with Smart Animate.
- Roaming toggle shows a confirmation overlay.
- Keypad buttons have pressed states.

## Expected Output

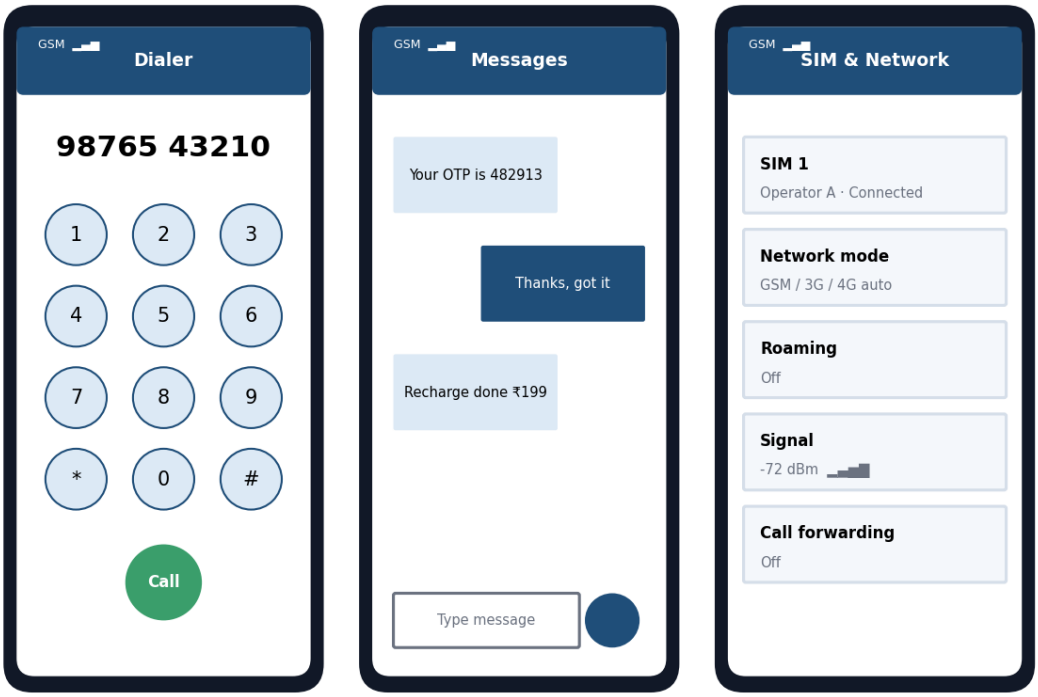



*Fig. 9.1 Reference layout to be recreated in Figma*

## Result

A six-screen interactive GSM service prototype was built in Figma with dialer, call, messaging and SIM/network settings screens.

## Viva Questions

- Name the main elements of the GSM architecture.
- What is the role of the SIM card?
- Which supplementary services are available in GSM?
- How is the roaming status shown to the user?